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1 Theoretic Foundations

1.1 Probabilistic Forecasting

Generally speaking, forecasts aim to predict a future event as accurately as possible. In classic deterministic forecasting one possible outcome is predicted and the discrepancy between forecast and realization - the prediction error - is measured and minimized. This approach suggests that the exact result can be predicted with a perfect prediction. However, information about future events is by definition incomplete and, thus, the events itself are necessarily subject to uncertainty. So, it can be beneficial to look at the problem from a probabilistic perspective, instead of ignoring the stochastic characteristic and minimizing the error of a deterministic prognosis. The goal of probabilistic forecasting is to model the uncertain behavior of the unknown event. This can be done by approximating its probability distribution or, in a more simple form, predicting confidence intervals or quantiles of the desired distribution. A popular example are weather forecast with rain probabilities (e.g. 80% chance of rain tomorrow). In general, meteorology has been and still is the driving field of applied probabilistic forecasting, see Murphy and Winkler [14], since weather forecasts always considered meteorological events part of probabilistic processes. Nowadays, probabilistic forecasting also finds applications in financial and economic risk managment [9], prediction of low-probability events like earthquakes [11] or floods [5].

In this section, we summarize the elementary theory of probabilistic forecasting based on the work of Gneiting and Katzfuss [8], who provides an overarching introduction to probabilistic forecasting. The section is structured into the theory of *prediction spaces*, the theoretic foundation, followed by the two most important concepts, *calibration and sharpness*, and finally concluded with *scoring* rules, the evaluation metrics for probabilistic forecasts.

1.1.1 Prediction Space

The theoretic framework for probabilistic forecasting is build on the idea of the *prediction space*, which was first introduced by Gneiting and Katzfuss [8] and developed further in Strähl and Ziegel [17]. The general problem is to predict the distribution for a random event. For our purpose sufficient, we discuss a simplified version of the prediction space based on two components.

First, we consider a one dimensional, real-valued random variable Y on the probability space (Ω, A, P) with a cummulative distribution function (CDF). Y is the future event, we aim to predict, and the probability space denotes the underlying probabilistic process. The second component is the prediction described by the random variable F on the same probability space. It takes a value in the space $\mathcal{F}$ of all continuous CDFs and measurable to the sub σ -algebra $A' \subseteq A$, hence $\mathcal{F} = \{F \text{ CDF} : F \text{ continuous} \wedge F \text{ } A'\text{-measurable}\}$. Thus, F is a random CDF depending on the information set A' , which can be interpreted as the

information available to the predictor like historic data or expertise. We will use the naming 'predicted distribution' for the CDF-valued random quantity F further on.

The event Y and prediction F are connected in the sense that the predictor constructs F in such a way with the available information A' that it represents the unknown distribution of Y as closely as possible. Formalizing this idea, we introduce the phrase *relative ideal* in the following definition in which $\mathcal{L}$ denotes the conditional distribution.

Definition 1.1. The predicted distribution F is ideal relative to the information set A' if $F = \mathcal{L}(Y|A')$

1.1.2 Calibration and Sharpness

In this section, we discuss beneficial properties of our predictive distribution F . Since in practice, it is not possible to observe the conditional distribution $\mathcal{L}(Y|A')$, we require different forms of assessment for the predicted distribution.

The first main idea is that the future event should occur with the same probabilities as predicted. If we could repeat the same experiment multiple times with a fixed forecast F , we would use the empiric distribution of Y and compare it to F . However in general, a repetition is not possible since for each time instance the available information and thus also F changes. To overcome this problem, the *probability integral transform* (PIT) is introduced¹.

Definition 1.2. The random variable $F(Y)$ is called the PIT of the predicted distribution F .

The PIT is a transform of the original random variable in the interval $[0, 1]$ and is the value that the predictive CDF attains at the observed event. Assuming that F is the true distribution of Y , we see that the PIT $F(Y)$ follows a uniform distribution. Furthermore, if we repeat the experiment for a different time instance, the resulting PIT should still be uniformly distributed. This property is generalized by the concept of *calibration* and elaborated farther in the following definition and theorem.

Definition 1.3. Given the random variable Y and the predicted distribution F .

- (i) The predicted distribution F is called *marginally calibrated* if $\mathbb{E}[F(y)] = \mathcal{P}(Y \leq y)$.
- (ii) The predicted distribution F is called *probabilistically calibrated* if its PIT is uniformly distributed, thus $F(Y) \sim U(0, 1)$.

[Difference Marginally Calibrated vs probabilistically calibrated]

Theorem 1.4. If the predicted distribution F is ideal relative to the information set A' , then it is also marginally and probabilistically calibrated.

¹For the case of discrete random variables, the PIT is formulated similar to definition 1.2 with technical additions for handling discontinuities.

Proof. Since F is relative ideal to A' , we can describe it as $F(y) = \mathcal{P}(Y \leq y | A')$. It follows the proof for marginal calibration,

$$\mathbb{E}[F(y)] = \mathbb{E}[\mathcal{P}(Y \leq y | A')] = \mathbb{E}[\mathbb{E}[\mathbb{1}_{\{Y \leq y\}} | A']] = \mathbb{E}[\mathbb{1}_{\{Y \leq y\}}] = \mathcal{P}(Y \leq y).$$

Hereby, the characterization of the probability function via the expectancy of the indicator function $\mathbb{1}$, and the tower rule was applied.

For the probabilistic calibration, we need the fact that $F(\cdot)$ is CDF-valued, which implies a continuous, monotonically increasing, surjective function. Thus, there also exists a monotonic increasing, right inverse function F^{-1} , hence

$$F \circ F^{-1} = id. \quad (1.1)$$

Since $F(\cdot)$ is additionally strictly monotonic and injective almost surely, the right inverse F^{-1} is also almost surely left inverse,

$$F^{-1} \circ F = id \quad a.s. \quad (1.2)$$

It follows for $s \in [0, 1]$,

$$\begin{aligned} \mathcal{P}(F(Y) \leq s) &= \mathcal{P}(F^{-1}(F(Y)) \leq F^{-1}(s)) \\ &= \mathcal{P}(Y \leq F^{-1}(s)) \\ &= \mathbb{E}[\mathcal{P}(Y \leq F^{-1}(s) | A')] \\ &= \mathbb{E}[F(F^{-1}(s))] \\ &= \mathbb{E}[s] = s. \end{aligned}$$

Respectively, the equations are argued by the monotony of F^{-1} , equation (1.2), the tower rule, the marginal calibration of F and equation (1.1). Thus, we have shown that $F(Y)$ follows a uniform distribution. $\square$

The theorem 1.4 also holds for non-continuous random variables, which is shown in Brockwell [4]. This result provides us with a necessary, but not sufficient, condition to assess whether a prediction is possibly relatively ideal. That is, by checking the PIT for uniform distribution. There are several statistical tests to check uniformity, such as the Kolmogorov-Smirnov test or the Chi-squared test, more practical, however, is assess the PIT graphically. The standard method the histogram of the PIT of a sufficient many PIT samples². In example, each 10% interval (0-10%, 10%-20%,...) should occur in 10% of all cases, which implies that each bar of the PIT histogram should have roughly the same height at 10%. The advantage over statistical numeric test is that this approach provides additional information of the behavior of the prediction. Characteristic errors such as skewness and over-/underdispersion can be easily spotted in the PIT histogram, see exemplary figures 1.

As mentioned, a uniformly distributed PIT does not imply that the prediction F is ideal to the information set A' . It applies also for predictions which do not use all of the available information, but are still calibrated. In mathematical terms, predictions which are ideal

²Also, the auto-covariance function of PIT is regularly used to study historic dependencies of forecasts when using time series data.

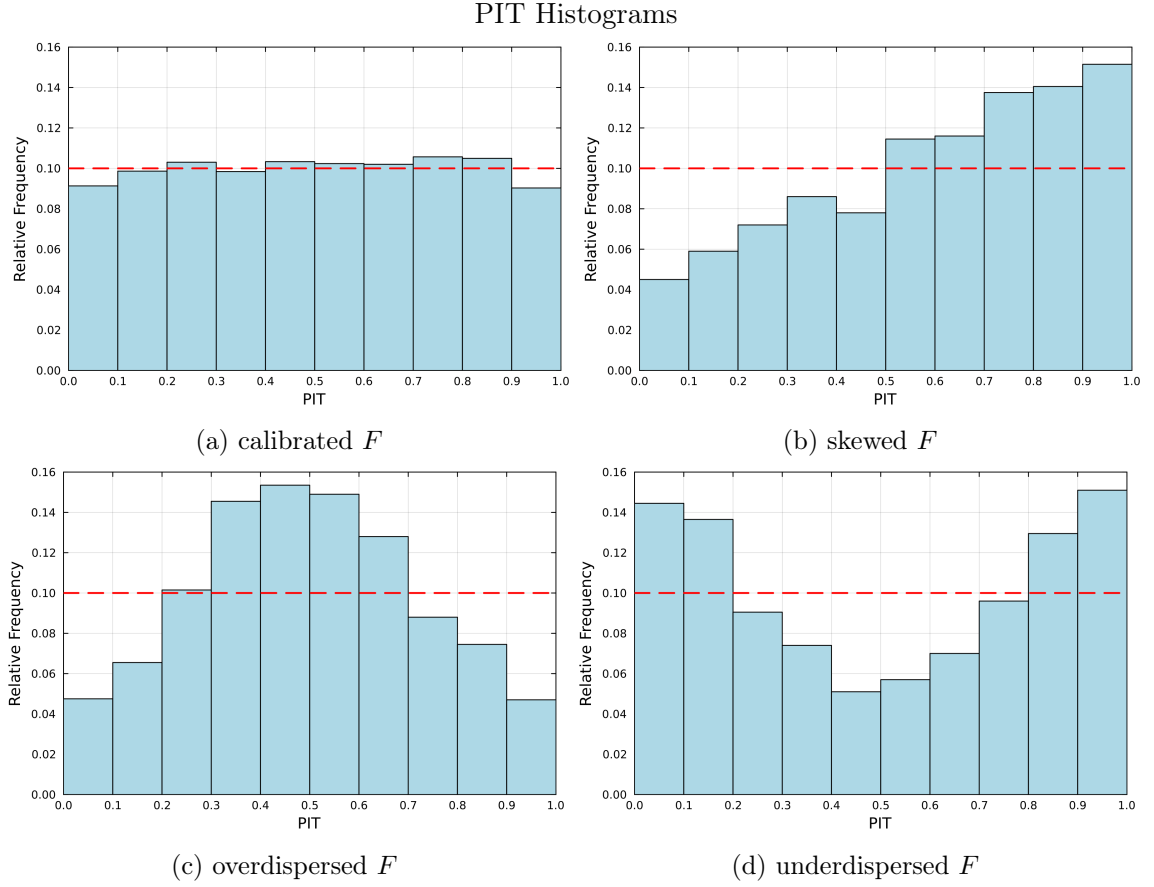


Figure 1: PIT histogram 1a shows a uniform distributed PIT. In figure 1b the higher intervals occur more often than predicted, while lower intervals occur less often, indicating that the predicted distribution is shifted to the right compared to the underlying distribution. The PIT of figure 1c does not obtain sufficiently often the extreme ends of the distribution implying the predictive distribution to be overdispersed. Whereas in figure 1d, the opposite is the case.

to other sub-sigma A^* algebras which are coarser than A' , thus $A^* \subset A'$. In example, the unconditional distribution of Y fulfills the condition of a uniformly distributed PIT and is also ideal to the trivial sub-sigma Algebra $A^* = \{\emptyset, \Omega\}$. However, it does not utilize any additional information given by A' . The question is, how can we assess that the maximum of available information was used?

The essential effect of utilizing more information is that uncertainty decreases, thus the predicted distribution becomes generally more concentrated. In probabilistic forecasting, the so-called attribute *sharpness* is used to describe the level of uncertainty. The "sharper" a distribution is, the less uncertain the predictive event is. Given now a set of calibrated predicted distribution, the prediction, which uses the most information, is also the one which maximizes the sharpness. Sharpness can be measured by known metrics like the variance,

confidence intervals, or the entropy of a distribution, for example the normal distribution $N(0, 1)$ is sharper than $N(0, 2)$. Notably, sharpness only takes the predicted distribution into account and does not consider the true realization of the future event. Thus, it evaluates exclusively the predicted distribution.

In summarizing this section, we present the dominating maxim of probabilistic forecasting formulated by Gneiting and Katzfuss [8]: "Probabilistic forecasting has the general goal of maximizing the sharpness of the predictive distributions, subject to calibration."

1.1.3 Scoring Rules and CRPS

This section discusses how probabilistic forecasts are evaluated numerically. It introduces the concept of *proper scoring rules* and gives a prominent example, the *continuous ranked probability score* (CRPS).

The evaluation metrics of probabilistic forecasts are not as straightforward as those applied in point forecasting. While in point forecasting, the distance between the point prediction and observed value is measured, the aim of the distribution forecast is to match the true underlying distribution, which, however, only "materializes" one observation per instance. Hence, the task is to measure the distance between distribution forecast and observation, and with sufficiently many evaluations, derive comparable results. These measures are called scoring rules, see [15] for a detailed approach.

Definition 1.5. The function $S : \mathcal{F} \times \mathbb{R} \rightarrow \mathbb{R}$ is called a scoring rule.

The scoring rule $S(\hat{F}, y)$ assigns the distribution $\hat{F}$ and the realization y of Y a numerical value. In order to be applicable for the evaluation of probabilistic forecasts, a scoring rule must be *proper*. Given a random variable Y , which follows the distribution $G \in \mathcal{F}$, the expression $S(\hat{F}, G)$ is denoted

$$S(\hat{F}, G) = \mathbb{E}_G[S(\hat{F}, Y)].$$

Definition 1.6. A scoring rule is *proper* if for all $\hat{F}, G \in \mathcal{F}$

$$S(G, G) \leq S(\hat{F}, G)$$

holds. It is *strictly proper* if $S(G, G) = S(\hat{F}, G)$ implies $\hat{F} = G$.

Thus, a proper scoring rule is designed in such a way that for the true underlying distribution the score is minimized in expectation.

In probabilistic load forecasting, the Continuous Ranked Probability Score (CRPS) has asserted itself in many applications, notably here is the Global Energy Forecasting Competition (GEFCom2014) summarized in Hong, Pinson, Fan, Zareipour, Troccoli, and Hyndman [10]. Its original form is defined by

$$\text{CRPS}(\hat{F}, y) = \int_{-\infty}^{\infty} (\hat{F}(x) - \mathbb{1}_{\{y < x\}})^2 dx.$$

The elementary idea is to calculate on the left side of the observation the area underneath the predicted distribution and on the right side the area above the predicted distribution,

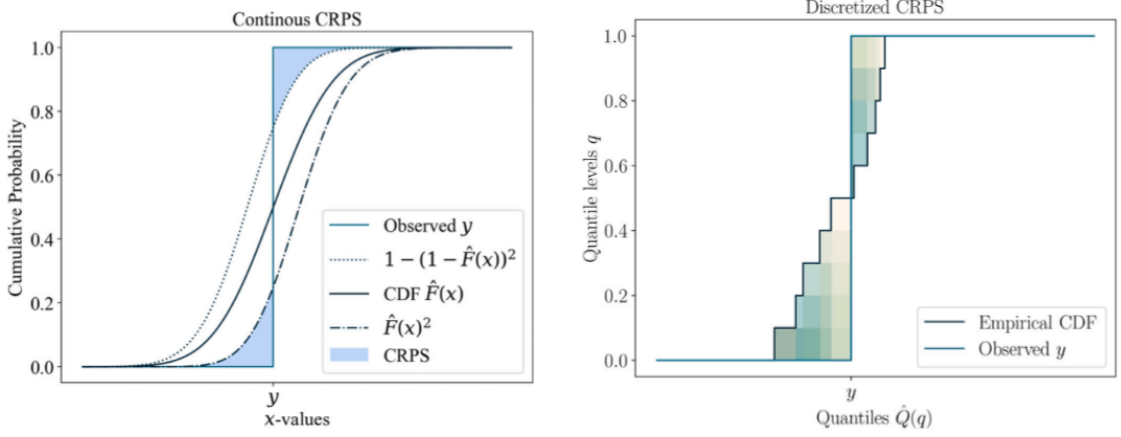


Figure 2: Graphical example of the continuous and discretized CRPS, reproduced from Botman, Lago, Becker, Vanthournout, and Moor [2]

see figure 2. The value is minimized, if the observation lies "central" and the distribution is as sharp as possible. In this way, the CRPS takes the two major concepts of probabilistic forecasting, calibration and sharpness, into account.

However, computing the continuous version of the CRPS can be computationally expensive. For this reason, we use a discretized version of the CRPS which is also applied for quantile predictions. This is formulated together with another popular scoring rule, the Pinball score,

$$\text{CRPS}(\hat{F}, y) = \sum_q \text{Pinball}(\hat{Q}(q), y, q)$$

with $\text{Pinball}(\hat{Q}(q), y, q)$:

$$= \begin{cases} (1 - q) \cdot (\hat{Q}(q) - y), & \text{if } y < \hat{Q}(q) \\ q \cdot (y - \hat{Q}(q)), & \text{if } y \geq \hat{Q}(q) \end{cases}$$

where q denotes a quantile and $\hat{Q}(q)$ the predicted value for q of the distribution prediction. For each evaluation, we calculate the quantiles 0, 0.01, 0.02, ... 0.99, 1 from our predicted distribution and compute the CRPS. To achieve comparable results, multiple evaluations are needed and then the mean CRPS is taken.

1.2 Hidden Markov Models

In simple words, a Hidden Markov Model (HMM) is a stochastic process, which underlies a hidden Markov process but can only be observed by probabilistic signals emitted from this hidden process. Originally, it was first studied by Baum and Petrie [1] as *probabilistic functions of Markov chains* and gained significance during the 1980s and 1990s as application in speech recognition, see Rabiner [16]. During this time, HMMs dominated automated speech recognition research and commercial application, forming the backbone of early voice-controlled interfaces. In general, HMMs are applied for tasks that require decoding, recognition and classification like DNA analysis and gene recognition [12, 18] or behavior detection [3]. Recently, Makonin, Popowich, Bajić, Gill, and Bartram [13] used HMM to disaggregate electric power load profiles and identify the usage of single electric devices. In the last years however, machine learning techniques like recurrent neural networks or transformers have superseded HMMs especially in large-scale tasks. Nonetheless, HMM still remains a valuable tool in many applications due to its interpretability, mathematical enclosed design and efficiency in low-data settings.

In this thesis, our aim is to investigate another possible application of HMM, namely forecasting. In the existing literature, there have been only a handful of studies conducted using HMM in forecasting [6, 7, 19]. However, to the best of the author’s knowledge, these studies have only been applied for deterministic predictions and for small-scale models (less than 10 hidden states). Applying HMMs for probabilistic forecasting offers a great opportunity because one of the biggest challenges in probabilistic forecasting is its computational effort, while HMMs provide probabilistic forecast very efficiently due to its stochastic design. Additionally, with today’s computer, models up to a couple of hundred hidden states are possible to calculate.

For this reason, we present in this section a theoretic approach to generate probabilistic forecasts using HMM and discuss the necessary theory and existing algorithms.

1.2.1 HMM Foundations

First off, we start with the notion of HMM. There are N possible hidden states forming the *state space* S and M different observation values forming the *observation space* O ,

$$\begin{aligned} S &= \{s_1, s_2, \dots, s_N\} \\ O &= \{o_1, o_2, \dots, o_M\}. \end{aligned}$$

We restrict the discussion to discrete HMM with finite state and observation space, thus $N, M \in \mathbb{N}$. Denote the index sets by $I_S = \{1, \dots, N\}$ and $I_O = \{1, \dots, M\}$. The HMM consists of two stochastic processes, a hidden and an observable one. For each time instance, the hidden process $\mathbf{Z}_T$ takes on values in the state space, while the observable process $\mathbf{X}_T$ is located in the observation space,

$$\begin{aligned} \mathbf{Z}_T &= (Z_1, Z_2, \dots, Z_T) \in S^T \\ \mathbf{X}_T &= (X_1, X_2, \dots, X_T) \in O^T, \end{aligned}$$

in which $T \in \mathbb{N}$ describes the time horizon. To distinguish between random variables and the event of occurring random variables, we introduce the following notation for better readability. The event of Z_t occurring is denoted by $\dot{Z}_t$,

$$\dot{Z}_t = \{Z_t = s_{i_t}\} \quad \text{for a fixed } i_t \in I_S,$$

and all events until time instance t summed up by the event vector $\dot{\mathbf{Z}}_t$,

$$\dot{\mathbf{Z}}_t = (\dot{Z}_1, \dot{Z}_2, \dots, \dot{Z}_t)$$

Analogously, for the observable process

$$\begin{aligned} \dot{X}_t &= \{X_t = o_{k_t}\} \quad \text{for a fixed } k_t \in I_O, \\ \dot{\mathbf{X}}_t &= (\dot{X}_1, \dot{X}_2, \dots, \dot{X}_t). \end{aligned}$$

The HMM is entirely characterized through the following three properties.

(A1) Markov property:

$$\mathcal{P}(\dot{Z}_{t+1} | \dot{\mathbf{Z}}_t) = \mathcal{P}(\dot{Z}_{t+1} | \dot{Z}_t)$$

(A2) Homogeneous Markov chain:

$$\mathcal{P}(Z_{t+1} = s_j | Z_t = s_i) = \mathcal{P}(Z_2 = s_j | Z_1 = s_i)$$

(A3) Independent observation assumption:

$$\mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{t-1}, \dot{X}_{t+1}, \dots, \dot{X}_T, \dot{\mathbf{Z}}_T) = \mathcal{P}(\dot{X}_t | \dot{Z}_t)$$

The properties **A1** and **A2** determine the process Z to be a homogeneous Markov chain. Thus, an event of the future depends on the past only through the present. The third property **A3** implies that the observation is influenced only by the current hidden state. Due to these properties, an HMM can be parameterized by stochastic matrices and vectors similar to the transition matrix of standard Markov chains. The hidden Markov process Z is determined by the $N \times N$ -dimensional *transition matrix* $\mathbf{A}$ with elements a_{ij} ,

$$a_{ij} = \mathcal{P}(Z_t = s_j | Z_{t-1} = s_i) \quad \text{for } i, j = 1, \dots, N,$$

and by the N -dimensional *initial distribution vector* π with

$$\pi_i = \mathcal{P}(Z_1 = s_i) \quad \text{for } i = 1, \dots, N.$$

The $N \times M$ -dimensional *observation matrix* $\mathbf{B}$ describes the probability of each observation conditioned by the hidden states with elements b_{ij} ,

$$b_{jk} = \mathcal{P}(X_t = o_k | Z_t = s_i) \quad \text{for } j = 1, \dots, N \text{ and } k = 1, \dots, M.$$

These three parameters fully describe a finite HMM, summarized by $\lambda = (\mathbf{A}, \mathbf{B}, \pi)$.

1.2.2 Algorithms

The three strong properties **A1**, **A2**, **A3** allow us to compute HMM related problems very efficiently. Before we present common algorithms, we identify first the three main questions regarding HMM for formulated by Rabiner [16]:

1. Evaluation:

Given an HMM and observation sequence $\dot{\mathbf{X}}_{\mathbf{T}}$, what is the probability that this sequence occurred,

$$\mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}})?$$

2. Decoding:

Given an HMM and observation sequence $\dot{\mathbf{X}}_{\mathbf{T}}$, what could be the optimal state sequence $\dot{\mathbf{Z}}_{\mathbf{T}}$ to describe the given observations,

$$\arg \max_{\dot{\mathbf{Z}}_{\mathbf{T}}} \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}} | \dot{\mathbf{Z}}_{\mathbf{T}})?$$

3. Training:

Given observation sequence $\dot{\mathbf{X}}_{\mathbf{T}}$, what are the optimal parameters λ of the HMM to describe the given observations,

$$\arg \max_{\lambda} \mathcal{P}_{\lambda}(\dot{\mathbf{X}}_{\mathbf{T}})?$$

Due to the Markovian behavior of HMMs, all of these quantities can be efficiently computed. In this section, we discuss solutions and algorithms for the problem of evaluation and training, which is relevant for the application of HMM as a forecasting tool. The question of decoding can also be relevant for the analysis of existing forecasting models to develop a deeper understanding of the underlying mechanics, which however we do not discuss further in this thesis.

To show the computational complexity of the above problems, we first make an exemplary naive approach of calculating the evaluation problem. Given $\dot{\mathbf{X}}_{\mathbf{T}}$, we reformulate the desired probability using the conditional probability over all possible hidden state sequences $\mathcal{Z}_{\mathcal{T}}$,

$$\begin{aligned} \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}}) &= \sum_{\dot{\mathbf{Z}}_{\mathbf{T}} \in \mathcal{Z}_{\mathcal{T}}} \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}}, \dot{\mathbf{Z}}_{\mathbf{T}}) \\ &= \sum_{\dot{\mathbf{Z}}_{\mathbf{T}} \in \mathcal{Z}_{\mathcal{T}}} \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}} | \dot{\mathbf{Z}}_{\mathbf{T}}) \mathcal{P}(\dot{\mathbf{Z}}_{\mathbf{T}}). \\ &= \sum_{\dot{\mathbf{Z}}_{\mathbf{T}} \in \mathcal{Z}_{\mathcal{T}}} \left(\prod_{t=1}^T \mathcal{P}(\dot{X}_t | \dot{Z}_t) \right) \left(\mathcal{P}(\dot{Z}_1) \prod_{t=2}^T \mathcal{P}(\dot{Z}_t | \dot{Z}_{t-1}) \right) \\ &= \sum_{\dot{\mathbf{Z}}_{\mathbf{T}} \in \mathcal{Z}_{\mathcal{T}}} \left(\prod_{t=1}^T b_{i_t k_t} \right) \left(\pi_{i_1} \prod_{t=2}^T a_{i_{t-1} i_t} \right). \end{aligned}$$

The set of all possible hidden state sequences is defined formally by

$$\mathcal{Z}_T = \{(\{Z_t = s_{i_t}\})_{t \in \{1, \dots, T\}} \text{ for } (i_t)_{t \in \{1, \dots, T\}} \in I_S^T\}.$$

Since I_S has N elements, there are N^T different possible hidden state sequences. Thus, evaluating $\mathcal{P}(\dot{\mathbf{X}}_T)$ has a computational complexity of order $\mathcal{O}(N^T \cdot 2T)$ and for larger models or longer time horizon reaches quickly the limits of computational feasibility. However, due to its Markovian behavior, the evaluation of $\mathcal{P}(\dot{\mathbf{X}}_T)$ can be done more efficiently using the so-called *forward algorithm*. It reduces the problem to an iterative procedure based on the Markov property shown in the following idea.

Assume $t \in 1, \dots, T$ and $i \in I_S$, then define

$$\alpha_t(i) = \mathcal{P}(\dot{\mathbf{X}}_t, Z_t = s_i).$$

For $t \geq 2$, the joint probability $\alpha_t(i)$ can be represented by its predecessors $\alpha_{t-1}(\cdot)$ and the parameters of the given HMM,

$$\begin{aligned} \alpha_t(i) &= \mathcal{P}(\dot{\mathbf{X}}_t, Z_t = s_i) \\ &= \sum_{j \in I_S} \mathcal{P}(\dot{\mathbf{X}}_t, Z_t = s_i, Z_{t-1} = s_j) \\ &= \sum_{j \in I_S} \mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{t-1}, Z_t = s_i, Z_{t-1} = s_j) \mathcal{P}(Z_t = s_i | \dot{\mathbf{X}}_{t-1}, Z_{t-1} = s_j) \\ &\quad \cdot \mathcal{P}(\dot{\mathbf{X}}_{t-1}, Z_{t-1} = s_j) \\ &= \sum_{j \in I_S} \mathcal{P}(\dot{X}_t | Z_t = s_i) \mathcal{P}(Z_t = s_i | Z_{t-1} = s_j) \mathcal{P}(\dot{\mathbf{X}}_{t-1}, Z_{t-1} = s_j) \\ &= \sum_{j \in I_S} b_{ik_t} \cdot a_{ji} \cdot \alpha_{t-1}(j). \end{aligned}$$

For the first elements, $t = 1$, the initial distribution π is needed,

$$\alpha_1(i) = \mathcal{P}(\dot{X}_1, Z_1 = s_i) = \mathcal{P}(\dot{X}_1 | Z_1 = s_i) \mathcal{P}(Z_1 = s_i) = b_{ik_1} \cdot \pi_i.$$

Using the quantity α , the evaluation problem is simply calculated by

$$\mathcal{P}(\dot{\mathbf{X}}_T) = \sum_{i \in I_S} \mathcal{P}(\dot{\mathbf{X}}_T, Z_T = s_i) = \sum_{i \in I_S} \alpha_T(i).$$

This is the standard forward algorithm, discussed in [16], and has a computational complexity of order $\mathcal{O}(N^2 \cdot 3T)$, which is significantly lower than in the naive approach. However, this algorithm also has its drawbacks in practice. If we look closely at the evaluation problem $\mathcal{P}(\dot{\mathbf{X}}_T)$, it is the case that with increasing time horizon T more and more probabilistic events are added, implying that the total probability is shrinking and in regular cases converging exponentially to 0. With a significantly large time horizon, this quantity will be smaller than any for computer representable number and thus not feasible. For this reason, we present a adapted version of the forward algorithm, which leads to the

feasible computation of the log-likelihood of the evaluation problem. We define two quantities $\tilde{\alpha}$ and $\bar{\alpha}$ similar to the original α ,

$$\tilde{\alpha}_t(i) = \mathcal{P}(Z_t = s_i | \dot{\mathbf{X}}_{\mathbf{t}})$$

and

$$\bar{\alpha}_t(i) = \mathcal{P}(\dot{X}_t, Z_t = s_i | \dot{\mathbf{X}}_{\mathbf{t}-1}).$$

On the one hand, $\tilde{\alpha}_t(i)$ can be expressed by $\bar{\alpha}_t(\cdot)$,

$$\begin{aligned} \tilde{\alpha}_t(i) &= \mathcal{P}(Z_t = s_i | \dot{\mathbf{X}}_{\mathbf{t}}) \\ &= \frac{\mathcal{P}(\dot{X}_t, Z_t = s_i | \dot{\mathbf{X}}_{\mathbf{t}-t})}{\mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{\mathbf{t}-t})} \\ &= \frac{\mathcal{P}(\dot{X}_t, Z_t = s_i | \dot{\mathbf{X}}_{\mathbf{t}-t})}{\sum_{j \in I_S} \mathcal{P}(Z_t = s_j, \dot{X}_t | \dot{\mathbf{X}}_{\mathbf{t}-t})} \\ &= \frac{\bar{\alpha}_t(i)}{\sum_{j \in I_S} \bar{\alpha}_j(t)}. \end{aligned}$$

On the other hand, $\bar{\alpha}_t(i)$ is computed by the predecessors $\tilde{\alpha}_{t-1}(\cdot)$,

$$\begin{aligned} \bar{\alpha}_t(i) &= \mathcal{P}(\dot{X}_t, Z_t = s_i | \dot{\mathbf{X}}_{\mathbf{t}-1}) \\ &= \sum_{j \in I_S} \mathcal{P}(\dot{X}_t, Z_t = s_i, Z_{t-1} = s_j | \dot{\mathbf{X}}_{\mathbf{t}-1}) \\ &= \sum_{j \in I_S} \mathcal{P}(\dot{X}_t, Z_t = s_i | \dot{\mathbf{X}}_{\mathbf{t}-1}, Z_{t-1} = s_j) \mathcal{P}(\mathbf{Z}_{\mathbf{t}-1} = \mathbf{s}_j | \dot{\mathbf{X}}_{\mathbf{t}-1}) \\ &= \sum_{j \in I_S} b_{ik_t} \cdot a_{ji} \cdot \tilde{\alpha}_{t-1}(j). \end{aligned}$$

Furthermore, for initializing the iterative computation,

$$\bar{\alpha}_1(i) = \mathcal{P}(\dot{X}_1, Z_1 = s_i) = \alpha_1(i) = b_{ik_1} \cdot \pi_i.$$

As mentioned, the probability $\mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}})$ induces underflow errors. However, the log-likelihood of the evaluation problem, $\log \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}})$ lies generally within the computationally feasible range. The loglikelihood is calculated by splitting up the original problem into subproblems,

$$\log \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}}) = \log \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{T}}) = \log \prod_{t=1}^T \mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{\mathbf{t}-1}) = \sum_{t=1}^T \log \mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{\mathbf{t}-1}),$$

and using the fact that $\mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{\mathbf{t}-t}) = \sum_{j \in I_S} \bar{\alpha}_j(t)$ as shown above. Additionally, consider

the following relation between $\tilde{\alpha}$ and α ,

$$\tilde{\alpha}_t(i) = \mathcal{P}(Z_t = s_i | \dot{\mathbf{X}}_{\mathbf{t}}) \quad (1.3)$$

$$= \frac{\mathcal{P}(Z_t = s_i, \dot{\mathbf{X}}_{\mathbf{t}})}{\mathcal{P}(\dot{\mathbf{X}}_{\mathbf{t}})} \quad (1.4)$$

$$= \frac{\mathcal{P}(\dot{\mathbf{X}}_{\mathbf{t}}, Z_t = s_i)}{\sum_{j \in I_S} \mathcal{P}(\dot{\mathbf{X}}_{\mathbf{t}}, Z_t = s_j)} \quad (1.5)$$

$$= \frac{\alpha_t(i)}{\sum_{j \in I_S} \alpha_j(t)}. \quad (1.6)$$

This shows that $\tilde{\alpha}_t(i) = c_t \cdot \alpha_t(i)$ with the constant factor $c_t = \frac{1}{\sum_{j \in I_S} \alpha_j(t)}$, thus $\tilde{\alpha}_t$ is a scaled version of α_t , which becomes relevant in the training problem.

Adapted forward algorithm

1. *Initialization:* ($t = 1$)

$$\bar{\alpha}_1(i) = b_{ik_1} \cdot \pi_i \quad \text{for } i \in I_S$$

$$\mathcal{P}(\dot{X}_1) = \sum_{j \in I_S} \bar{\alpha}_1(j)$$

$$\tilde{\alpha}_1(i) = \bar{\alpha}_1(i) / \mathcal{P}(\dot{X}_1) \quad \text{for } i \in I_S$$
2. *Iteration:* ($t = 2, \dots, T$)

$$\bar{\alpha}_t(i) = \sum_{j \in I_S} b_{ik_t} \cdot a_{ji} \cdot \tilde{\alpha}_{t-1}(j) \quad \text{for } i \in I_S$$

$$\mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{\mathbf{t}-1}) = \sum_{j \in I_S} \bar{\alpha}_t(j)$$

$$\tilde{\alpha}_t(i) = \bar{\alpha}_t(i) / \mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{\mathbf{t}-1}) \quad \text{for } i \in I_S$$
3. *Termination:*

$$\log P(\vec{X}) = \sum_{t=1}^T \log \mathcal{P}(\dot{X}_t | \dot{\mathbf{X}}_{\mathbf{t}-1})$$

return $\log P(\vec{X})$

[Evtl. Problembeschreibung hierherverschieben]
- herleitung von p - Forward Algo - Computational Novel..., skalierung - Backward Algo -
skalierung... - Allgemein EM Algorithms - Herleitung Baum welch - Baum Welch

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